

Exact Bloch Oscillations and Schatten Boundaries for the Wannier–Stark Lattice

Route-A source-local certificate HCS-C267

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Abstract

For the uniform-field tight-binding Hamiltonian on $\ell^2(\mathbb{Z})$ we give one all-parameter closure: a periodic Fourier gauge, the simple Wannier–Stark ladder and Bessel basis, the exact propagator and least identity return. We also derive the delta-source probability shell and second moment, including the zero-hopping and changed-owner zero-field faces. Finally we separate compact resolvent from noncompact unitary dynamics and prove the sharp resolvent Schatten threshold. No infinite-dimensional conclusion is inferred from finite truncation.

1 Frozen owner and unitary normal form

For $F, J \in \mathbb{R}$, $F \neq 0$, set

$$(H\psi)_n = Fn\psi_n - J(\psi_{n+1} + \psi_{n-1}) \quad (1)$$

on the domain of multiplication by n in $\ell^2(\mathbb{Z})$. This is the one-band uniform-field lattice associated with the Wannier–Stark problem [1]; exact lattice propagators in this setting were studied by Yellin [2]. These citations establish lineage, not novelty.

Fix the convention $(\mathcal{F}\psi)(k) = \sum_{n \in \mathbb{Z}} \psi_n e^{ink}$ into $L^2(\mathbb{T}, dk/2\pi)$. Then

$$\hat{H} = -iF\partial_k - 2J \cos k.$$

Let G multiply by $\exp[-i(2J/F) \sin k]$. Direct differentiation gives

$$\hat{H} = G^{-1}(-iF\partial_k)G. \quad (2)$$

The sign in (2) is part of the contract: reversing it reverses the hopping.

Theorem 1 (ladder and basis). *H has simple pure-point spectrum $F\mathbb{Z}$. A complete orthonormal eigenbasis is*

$$\phi_m(n) = J_{n-m}(2J/F), \quad H\phi_m = Fm\phi_m. \quad (3)$$

Proof. The circle momentum in (2) has the simple orthonormal eigenbasis e^{imk} . Applying G^{-1} and using $e^{ia \sin k} = \sum_r J_r(a) e^{irk}$ gives (3), including its index. Equivalently, the Bessel recurrence $J_{r-1}(a) + J_{r+1}(a) = 2rJ_r(a)/a$ verifies the eigen-equation. Gauge unitarity supplies completeness and orthonormality, rather than a finite-volume limit. $\square$

2 Exact evolution and least return

Theorem 2 (propagator). *For all $m, n \in \mathbb{Z}$ and $t \in \mathbb{R}$,*

$$U_{nm}(t) = i^{n-m} e^{-iFt(n+m)/2} J_{n-m}\left(\frac{4J}{F} \sin \frac{Ft}{2}\right). \quad (4)$$

The least positive T for which $U(T) = I$ on the full space is $T_B = 2\pi/|F|$.

Proof. Solving the Fourier transport equation along characteristics, or inserting (3) and applying the Bessel addition formula, yields (4). Its off-diagonal sentinel is $U_{m+1,m}(t) = +iJt + O(t^2)$, as required by $U(t) = I - itH + O(t^2)$. At T_B , the Bessel argument is zero and the diagonal phase is one, so $U(T_B) = I$. Conversely, $U(T) = I$ applied to the eigenvalues 0 and F forces $e^{-iFT} = 1$; hence no smaller positive return exists. $\square$

3 Probability shell, moment, and boundary atlas

For $\psi(0) = \delta_0$, put $z(t) = (4J/F) \sin(Ft/2)$. Formula (4) gives the exact shell

$$\mathbb{P}\{X_t = n\} = |U_{n0}(t)|^2 = J_n(z(t))^2. \quad (5)$$

The characteristic function of (5) is $J_0(2z \sin(q/2))$. Its value and first two derivatives at zero give

$$\sum_n \mathbb{P}\{X_t = n\} = 1, \quad \mathbb{E}X_t = 0, \quad \mathbb{E}X_t^2 = \frac{z(t)^2}{2} = \frac{8J^2}{F^2} \sin^2 \frac{Ft}{2}. \quad (6)$$

Thus spreading is bounded and returns exactly, not just asymptotically.

When $J = 0$, (1) is diagonal: (2)–(4) and the least return remain valid, while (5) is frozen at zero. The limit $F = 0$ with $J \neq 0$ changes owner. Fourier multiplication by $-2J \cos k$ has absolutely continuous spectrum $[-2|J|, 2|J|]$, kernel $i^{n-m} J_{n-m}(2Jt)$, and no positive full-space identity return. It is therefore not inserted into the $F \neq 0$ theorem.

4 Sharp operator-ideal boundary

Theorem 3 (unitary versus resolvent). *Every $U(t)$ is noncompact and belongs to no finite Schatten class. For $\lambda \notin F\mathbb{Z}$,*

$$(H - \lambda)^{-1} \in \mathcal{S}_p \iff p > 1.$$

In particular, the resolvent is compact and Hilbert–Schmidt but not trace class.

Proof. A unitary on an infinite-dimensional Hilbert space sends an orthonormal sequence to another orthonormal sequence, which has no norm-convergent subsequence; compactness fails. Every finite Schatten class is contained in the compact operators. By (3), the resolvent singular values are exactly $|Fm - \lambda|^{-1}$, $m \in \mathbb{Z}$. Their p th powers form a two-sided series comparable to $|F|^{-p} \sum_{m \neq 0} |m|^{-p}$, which converges exactly for $p > 1$. $\square$

5 Executable certificate and Route-A boundary

The deterministic receipt contains 210 signed parameter–time rows, 1,050 kernel cells, 5,250 shell cells, and 1,890 eigen cells. An implementation independent of the producer closes 12,335 assertions, including direct Schrödinger residuals that fix all phases and indices. SymPy closes 142 identities, byte replay is exact, and repaired-hash hostile testing rejects 20/20 changes. These are regression sentinels; equations (2)–(6) and the operator arguments above prove the infinite-dimensional claims.

The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`. This owner supplies a natural self-adjoint quantum Hamiltonian and an exact return clock, but no target arithmetic local data, Euler factors, root numbers, automorphy, target divisor, functional equation, or target Hilbert–Pólya identification. The strict tuple is

$$(\text{AO_FAIL}, \text{A1_WEAK}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_NATURAL_QUANTIZATION}).$$

The verdict is `ROUTE_A_REJECTED`; Route B is disabled.

References

- [1] G. H. Wannier, *Wave Functions and Effective Hamiltonian for Bloch Electrons in an Electric Field*, Phys. Rev. **117** (1960), 432, [doi:10.1103/PhysRev.117.432](https://doi.org/10.1103/PhysRev.117.432).
- [2] J. Yellin, *Two exact lattice propagators*, Phys. Rev. E **52** (1995), 2208, [doi:10.1103/PhysRevE.52.2208](https://doi.org/10.1103/PhysRevE.52.2208).